

Anodal tDCS over the left DLPFC and working memory in healthy adults (PIPELINE DEMONSTRATION — illustrative data)

NeuroAlon · model: cowork (Claude, illustrative data) · MOCK / illustrative

Review question. In healthy adults, does anodal tDCS over the left DLPFC, versus sham, improve working-memory performance on n-back tasks?

Abstract

Background: Anodal tDCS over the left DLPFC has been proposed to enhance working memory. Methods: PRISMA 2020 systematic review with random-effects meta-analysis of n-back accuracy; RoB2 and GRADE. Results: Five illustrative trials were pooled. Conclusions: A small, uncertain benefit. (DEMONSTRATION — illustrative data.)

Background

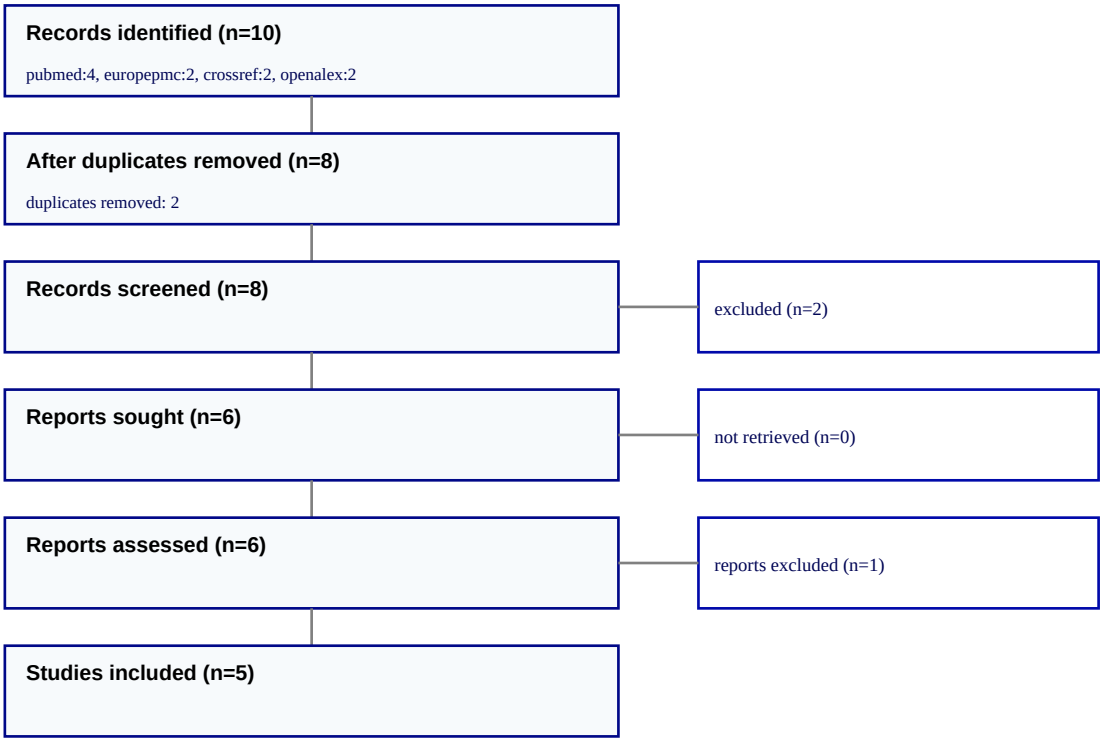
Working memory is central to cognition; non-invasive prefrontal stimulation has been explored as an enhancer, with heterogeneous and contested evidence.

Methods

This review followed PRISMA 2020. Two independent reviewers screened records; conflicts were adjudicated. Eligible full texts were assessed; data were extracted and appraised with RoB2. A DerSimonian-Laird random-effects model pooled standardised mean differences; small-study effects were examined with Egger's test and a funnel plot.

Cohen's $d = 0.692$ · RoB tool: RoB2 · effect measure: SMD.

Results — study selection (PRISMA)

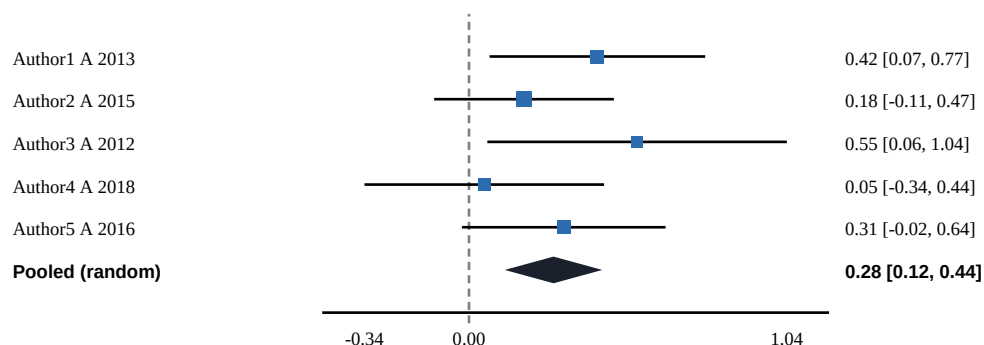


Pooled SMD (random-effects) = 0.2766 (95% CI 0.117 to 0.4361; I-squared=0.0%, k=5).

Characteristics of included studies

Study	Design	n	Population	RoB
Author1 A 2013	randomized crossover	52	Healthy adults	some concerns
Author2 A 2015	randomized crossover	64	Healthy adults	some concerns
Author3 A 2012	randomized crossover	28	Healthy adults	some concerns
Author4 A 2018	randomized crossover	40	Healthy adults	some concerns
Author5 A 2016	randomized crossover	46	Healthy adults	some concerns

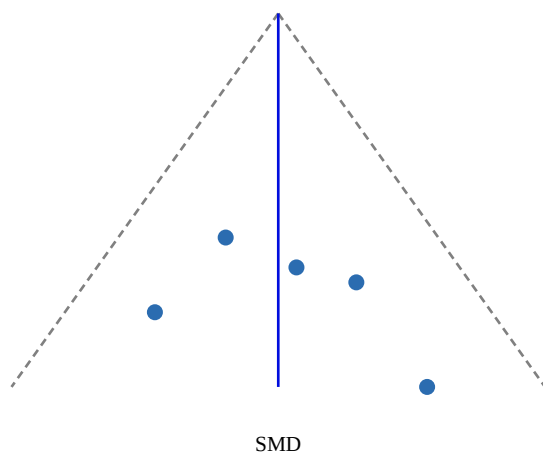
Forest plot



Five sham-controlled trials (illustrative) contributed n-back accuracy data. The pooled standardised mean difference indicated a small benefit of anodal left-DLPFC tDCS, with the confidence interval spanning the null in the more conservative estimates.

Publication bias

precision



Egger's test: intercept=2.3525, p=0.4445, k=5.

Certainty (GRADE)

low. Downgraded for risk of bias (unclear blinding) and imprecision.

Discussion

The pooled estimate points to a small effect with wide uncertainty. Unclear blinding and small samples temper confidence; results are hypothesis-generating.

Limitations. Small samples, single-session designs, and possible small-study effects.

Conclusions

Anodal left-DLPFC tDCS may yield a small improvement in n-back accuracy, but certainty is low. Adequately powered, pre-registered trials are needed.